[redacted]

A/B, VII, 3, 1

27 September 1951

MEMORANDUM FOR: Deputy Director for Central Intelligence

[redacted] G

SUBJECT : Operational Report on Project Artichoke (Formerly Bluebird)

1. Attached is an operational report on Project Artichoke which will be of interest to you.

2. This has been coordinated within the Agency and, as you will note from the forwarding memorandums attached to it, it is being sent to the service intelligence designees.

3. It is requested that this report be returned to [redacted] when you are finished with it. G

[redacted] A

Attach: Rpt on Project Artichoke

[redacted]

A/B, VII, 3, 2

[redacted]

MEMORANDUM FOR: [redacted] A

SUBJECT : Operational Query H-B/3

1. The attached report describes an incident of interest to the official community active in the Artichoke Project.

2. Through its own channels CIA will endeavor to obtain further information relating to the nature of material. In the meantime the services are cautioned to be on the alert for an opportunity to obtain further samples of the unknown. They are also asked to inform CIA whether or not a similar device, in the whole, or component parts has come into their possession.

[redacted] A

cc: [redacted] G

[redacted]

[redacted]

A/B, VII, 3, 3

SUMMARY OF REPORT

1. In July 1951 two [redacted] agents were seized in [redacted] F They carried tubes supposed to contain a material which when injected into persons would make the victim amenable to the will of the captor for an indefinite length of time.

2. One of the samples was given to a field laboratory which was unable to identify the material. Their test included many common drugs. The other sample was brought to America and given to a prominent drug firm for analysis. They were unable to determine the exact nature of the material, however, their findings somewhat limited the number of possibilities.

[redacted]

[redacted] A/B, VII, 3, 4

1. In July 1951 two [redacted] agents were seized in the [redacted] F [redacted] Each of them carried a clear plastic cylindrical container one and one-quarter inches in diameter and four and one quarter inches long. The container was divided into two sections, one of which was one and one-quarter inches long. The center of the circular partition dividing the two sections was perforated by a small hole. Other than this the smaller section had no opening. The larger section was closed at the end away from the partition by a friction lid the width of the tube diameter. In the larger section was a collapsible tube which had been equipped with a needle.

2. The agent stated that the collapsible tube contained a drug which, when injected into any part of the body, would render the victim completely amenable to guidance of his captor for an indefinite period of time. The victim could walk, stand erect, and would show no evidence of narcosis.

3. One of the cylinders was turned over to a field laboratory, and after a routine check of the common drugs, reported that they could not identify the material. The other cylinder, through [redacted] was turned over to an outside company whose facilities and accomplishments in the drug field are unsurpassed. This company examined the material, and have given a preliminary report as follows:

a. There is a certain amount of one-celled vegetable contamination.

b. There were large quantities of a material which was identified as calcium sulphate. Therefore the material is probably not a biological, and may be the calcium complex of an alkaloid.

[redacted]

c. Through spectrographic analysis they have determined that the material is not scopolamine; is definitely not sodium barbital, and probably is not a barbiturate.

They have recently completed their work attempting to find a matching absorption group and the final report is on its way to Washington.

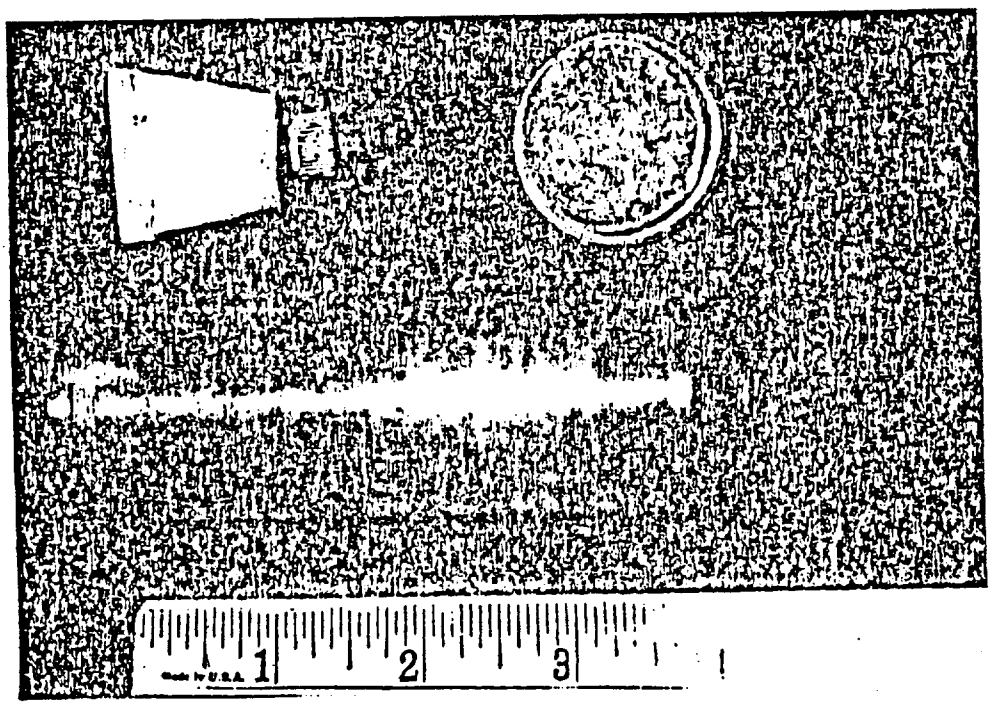
[redacted]

DETAILED DESCRIPTION

A/B, VII, 3, 5

1. The container was a clear plastic cylindrical container one and one-quarter inches in diameter and four and one-quarter inches long. This container was divided into two sections, one of which was one and one-quarter inches long. The center of the circular plate dividing the two sections was perforated by a small hole. Other than this hole the smaller section had no opening. The larger section was closed at the end away from the small perforation by a friction lid the width of the tube's diameter. In the larger section was a collapsible tube which had been equipped with a needle. Inside of the container were also some coffee colored spots which had a musty odor.

2. Inside was a collapsible tube supposed to contain a drug which, when injected into any part of the body, would render the victim completely amenable to the guidance of his captor for an indefinite period of time. The victim could walk, stand erect, and would show no evidence of narcosis.



[redacted]

EXAMINATION

A/B, VII, 3, 6

1. One cylinder was turned over to a field medical laboratory and the report of their analysis is below:

a. An unknown liquid, measuring approximately 1.5 cc in volume was personally delivered to the undersigned o/a 11 July 1951 with request for analysis.

b. Request complied with and results are as follows:

(1) Original volume - 1.5 cc
After centrifugation - 1.1 cc

(2) Reaction - slightly acid

(3) Fujiwara test for Chloral Hydrate and Chloroform - Negative (0.4 cc material used)

(4) The remaining fluid and the sediment were tested separately for the following:

(a) Barbiturates - Negative

(b) Salicylates - Negative

(c) Atropine, cocaine, codein, quinine, strychnine - Negative

(d) Morphine, theobromine, apomorphine - Negative.

2. The other sample was submitted to a drug company whose facilities and personnel are outstanding. Their report is as follows:

a. Examination of the tube under ordinary and under ultra-violet light shows no marking or label.

b. Identification of the material in the collapsible tube was impossible due to inadequate sample. The tube contained largely air. Existing material was dried out and had apparently undergone some decomposition, however, two droplets of liquid were extruded.

c. The following facts were developed as a result of certain standard procedures.

[redacted]

(1) Microscopic examination:

(a) Present was a large amount of one-celled vegetable material; probably contamination.

(b) A large quantity of crystals of calcium salt was present. This indicated the material was probably not a biological but might well be the calcium complex of a drug; such as an alkaloid.

(2) Ultra-Violet Absorbtion Specroscopic Examination with a Cary Recording Spectrophotometer

(a) Sodium Barbital and Scopolamine had such different curves that they could be eliminated as possibilities.

(b) The material yielded a characteristic "shoulder" at 290 milli-microns.

(c) This curve obtained was not similar to any known drug.

[redacted]

[redacted] : A/B, III, 3, 9.

[redacted]

REAGENTS: 1) One percent aqueous solution Benzidine Dihydrochloride, prepared by shaking the proper quantities of benzidine dihydrochloride and distilled water at intervals for about 2 hours, or until dissolved. Filter and keep in a dark amber bottle. Do not use heat in preparing the solution.

2) Three percent solution hydrogen peroxide, made by diluting C.P. 30% Hydrogen Peroxide with water in the proportion of 1 volume of the 30% Peroxide to 9 volumes of water.

3) One percent aqueous solution Reagent Sodium Acetate Crystals.

PROCEDURE: About 2½ mg. of debris and rust scraped from the hypodermic needle fragment were collected in a small centrifuge tube. About 5 ml. of distilled water were added and the suspension was agitated with a stirring rod for several minutes. The tube and contents were centrifuged for 15 minutes and the clear aqueous portion was adjusted to 5 ml. and transferred to a cylinder. To the color mixed, 1 ml. of benzidine dihydrochloride solution, 1 ml. of hydrogen peroxide solution and 1 ml. of sodium acetate solution were added. A blue color developed within 30 seconds. This test was performed with a control of fresh diluted human blood plasma treated in a similar manner. The control produced a similar blue color in approximately 10 seconds.

[redacted]

COPY

FINDING: [illegible]

REAGENTS:

1) One percent aqueous solution Benzidine Dihydrochloride, prepared by shaking the proper quantities of benzidine dihydrochloride and distilled water at intervals for about 2 hours, or until dissolved. Filter and keep in a dark amber bottle. Do not use heat in preparing the solution.

2) Three percent solution hydrogen peroxide, made by diluting C.P. 30% Hydrogen Peroxide with water in the proportion of 1 volume of the 30% Peroxide to 9 volumes of water.

3) One percent aqueous solution Reagent Sodium Acetate Crystals.

PROCEDURE: About 25 mg. of debris and rust were collected in a small centrifuge tube. About 5 ml. of distilled water were added and the suspension was agitated with a stirring rod for several minutes. The tube and contents were centrifuged for 15 minutes and the clear aqueous portion was adjusted to 5 ml. and transferred to a cylinder. In the order named, 1 ml. of benzidine dihydrochloride solution, 1 ml. of hydrogen peroxide solution and 1 ml. of sodium acetate solution were added. A blue color developed within 30 seconds. This test was performed with a control of fresh diluted human blood plasma treated in a similar manner. The control produced a similar blue color in approximately 10 seconds.

COPY

Copies

TO : [redacted]

FROM : [redacted]

19 September 1951

A/B, III, 3,10

SUBJECT: Vial

Reference is made to our discussion in the conference of Wednesday, 19 September 1951, concerning the instant case. For any help it may be to you, I am listing below the questions that occur to me in connection with this instrument. If this list is of no value, it can be burned.

Needle

1. Is the needle a standard medical needle designed for injection?
2. Assuming the dimensions of the vial to determine the size of the needle, can it be determined whether the needle can be used for venipuncture, or could the needle be used for general work such as intramuscular puncture.
3. Of what material is the needle made?
4. Is it a high-grade material or a sloppily produced job?
5. Could it possibly be used for animal injection?
6. What is the diameter of the bore?
7. If the bore is clogged, what is it clogged with--rust, blood, chemicals, etc.?
8. Is it the type of needle that only an experienced medical man would use?
9. Is the needle a standard medical instrument such as made in the big medical manufacturing houses?
10. Could the needle possibly be used by "scratching" such as used in common vaccination?
11. Is the bore of the needle of sufficient size so that a gelatinous substance could be forced through it or would it necessarily have to be a liquid?
12. Is it possible that any such needle could be used to plant crystals, dust, or powder in a human body?
13. Does the needle occur in any standard medical equipment or Army or Air Corps equipment or equipment of any related type?
14. Would a microscopic examination indicate it had ever been used on humans or animals, i.e. minute pieces of flesh at the puncture end or possibly hair?
15. Does the needle show any indication of being supported by a cork or plastic to hold it stiffly in place in the lower section?

The Collapsible Tube

1. Can medical research determine whether or not drugs, serums, etc. are commonly handled in collapsible tubes in the USSR or the satellite countries?

[redacted]

2. What is the tube made of?
3. Can the tube stand heavy pressure?
   (a) Could it be filled by squirting a liquid or a jelly into it?
   (b) Could it be refillable?
   (c) Could it be used by a person carelessly in a rough or sloppy manner without damage to the tube? In short, how strong is the tube and what are its characteristics?
4. Were the folds at the top of the tube made by hand or by machine, or possibly by a special instrument?
5. Does a microscopic examination of the outside of the tube indicate that it was packed in cotton or some soft material to prevent rough treatment?
6. What is the best guess as to what the tube contained? Gelatin, liquid, powder, etc.?
7. Could the two drops that were extracted from the tube be a residue after the liquid evaporated?
8. Is calcium sulphate ever the basis for a poison, a hypnotic, or chemical used in treatment of humans or animals?
9. Is the tube fireproof or heatproof?
10. Could the entire contents of the tube be squirted through the needle in one thumb and forefinger squeeze?
11. Is there any indication anywhere of dosages required?
12. Is the small nipplelike mechanism in the mouth of the tube a type that would hold back a water solution, an alcohol solution, a powder, a gelatin solution, etc.?
13. From a medical point of view, could the device be used to give a quick injection, i.e. is it so designed?
14. Was the original [redacted] report made by the [redacted] toxicologist ever been given to this agency? Can it be obtained? Any why can't it be turned over to [redacted] or other competent people for their study?
15. Why can't the [redacted] people or other low-level scientists submit a series of questions to the [redacted] toxicologist, re the material in the tube, for him to answer?
16. Were the contents of the other tube (as well as all component parts) saved? If so, why can't they be brought to the United States for proper study?

H-B-3, F
C
C
F

The Container

1. Why is the container shaped as it is?
2. As it is, it is a bulky, unwieldy object that could not be easily concealed on the person or for that matter hidden inconspicuously in many places. It is obvious that the tube is not designed for concealment or to represent what it is but what it is.
3. Is the container fireproof?
4. Is the container, when both ends are on, waterproof and heatproof?
5. Why was the container built in such a heavy manner?

6. What does a scanning of the inside of the container show under chemical analysis?
7. Is there any evidence that cobalt or some other material was used to support the collapsible tube?
8. Is there any evidence to indicate that instructions were pasted on the inside of the container or on the outside or any of the component parts thereof?
9. Has the container (and for that matter, the entire instrument) ever been seen before?
   (a) Pictures of this should be sent to any agency or individual having working knowledge of [redacted] equipment for possible identification.
10. Is the container similar to any standard [redacted] container by outside dimensions, material, color, design, etc.?
11. Is the container sufficiently large to hold two of the collapsible tubes?
12. Would it be possible to find out what instructions were given to the [redacted] toxicologist in the instant case?
   (a) I note that it was turned over to the toxicologist for "a routine check of the common drugs". It would be interesting to know in this case what was meant by common drugs.

H

F

The [redacted] Agents

H

1. Were they ever run on the polygraph?
2. Is there a possibility that they were drug addicts? Are there any needle-puncture marks on the arms of either of the agents? Have they had a psychiatric examination?
3. Were these agents of high, medium, or low caliber intellectually?
4. Were tests made to see whether or not the agents could administer a substance by the use of a needle?
5. Was a verbatim written report of the agents' interrogation ever submitted to this agency?
6. If these agents were low-grade, or even medium-grade operators, why would they be entrusted with such a valuable weapon?
7. Have we ever received information of a similar situation, or is there any other report that would indicate Soviets have ever carried hypodermic equipment of any type, or has there ever been any instruction to Soviets on how to give hypodermics?
8. Did the agents themselves ever use the instrument? If so, where, when, why, results, how used, how much given, etc., etc.?
9. Was the instrument carried on the person of the agent or in a bag with him, or was it carried in some concealed manner?
10. If the drug is represented as being so powerful, was it used on the agents themselves? If not, why not?
11. Have the agents ever seen tests made in the agent schools or read any reports on the use of this weapon?

[redacted]

12. Does the interrogation report indicate whether or not only one instrument was issued per agent, or were the instruments packed in a group in a special type of container that would perhaps contain ten or twelve of the instruments?

General

1. Can the [redacted] report in this matter be obtained through liaison channels? H
2. As stated above, can we obtain any residue material left after the toxicologist's work, or even the other container or parts thereof?
   (a) This might be valuable to determine whether or not by microscopic examination if they were built on the same machines or built by the same medical instrument manufacturing company or plastic manufacturing company.

   It appears obvious that this weapon, if it is as good as stated, would be extensively used and certainly there would be rumors, gossip, or intelligence information concerning this.
3. Why would the [redacted] make a "field-laboratory" examination of this matter and not send it immediately to the best research people in the United States for study? H
4. Did the agent state how much of the material in the tube was necessary to get control of an individual--all, half, one drop, or any given part?
   (a) When the agents were seized, specifically what was the condition of both instruments?
   (b) Were photographs taken of same at this time?
   (c) Was the container weighed at this time?
5. Could the entire situation be a rigged job? If so, why such a crude, bulky design, or would that be part of the rig?
6. Is the collapsible tube designed to hold liquid, jelly, or powder under pressure so that if squeezed it would cause same to squirt out with considerable force?
7. Could the two drops that were found be a means of holding another solution or a biological in a solution of some sort.
   (a) Although [redacted] rules out the biological angle, could not the biological disappear upon exposure to air and leave no trace. In connection with this, it is suggested that [redacted] might possibly have valuable information. C H
8. Is there any information that would indicate that germs or germ produced disease could be used or had been used as a hypothetic? In this connection, very fast acting diseases possibly producing very high temperatures are certain to have been studied by the [redacted] groups. C

For a matter of record, our research does not show any literature along these lines, but it is known that in some individuals, serums or anti-toxins will produce delerium, hallucinations, etc.

[redacted] A

[redacted]

9 January 1952

A/B, VII, 3, 11

SUBJECT: Second Report on Item "Plastic"

1. On 26 September 1951 a preliminary report on an object, which has come to be known as Item "Plastic", was circulated to CIA and [redacted] personnel concerned with Project Articboke. (Copy attached) -H

2. The preliminary report gave a brief description of the item and the manner in which it was obtained. A picture of the components was included and it was indicated that analysis of the contents of the tube both in the field and in the United States failed to identify the material.

3. The individuals from whom the item was obtained claimed that the tube contained a material which when injected in a victim, would make him amenable to the will of his captor. Therefore evaluation efforts have been made through chemical analysis of the item to determine the true nature and purpose of the material it contained. To date these efforts have been unsuccessful. However, information obtained since the original report is summarized herein for record purposes:

a. One of the containers was retained in the field and tested for barbiturates with negative results. No information on the extent of testing and the disposition of the material has been obtained. Further inquiry has been made and a highly competent scientific consultant is available to follow up on any leads should they develop.

b. Information has been obtained indicating that originally the tube was full of a jelly-like substance. On delivery for chemical testing, however, the tube was practically empty and complete analysis of the contents was therefore impossible.

c. Despite this factor two drops of the material were obtained from the tube and subjected to spectrophotometric analysis. The resultant curve had a peak shoulder at 290 milli-microns which is not similar to that exhibited by any substance or compound known to the investigator. Efforts to synthesize a compound with similar characteristics have been unsuccessful to date. Scrapings of the inside of the container have been subjected to exhaustive analysis with no positive results to date.

d. Scrapings of rust and decomposed matter in the order of 10 to 25 milligrams from the outside surface of the needle gave a positive test for blood with the Benzidine test. This is not conclusive evidence that there was blood on the needle and further information on the manner in which the test was performed has been requested. If this fact is established it is significant because of the agent's statement that the item had not been used.

[redacted]